

Replication Report: Brogi et al. (2016)

Rotation and Winds of Exoplanet HD 189733b from High-Resolution Spectroscopy

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Abstract

This report details the full replication of Figures 1 and 2 from Brogi et al. (2016) (*ApJ*, 817, 106) using our `hot_jupiter` C++ core library (`Brogi2016WindRotationModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9988$) across day-to-night jetstream wind blueshifts ($v_{\text{wind}} = -1.9 \text{ km s}^{-1}$) and planetary rotational broadening ($v_{\text{rot}} \sin i = 3.4 \text{ km s}^{-1}$).

1 Introduction & Methodology

Brogi et al. (2016) resolve high-resolution Doppler wind blueshifts and rotational line-broadening kernels:

$$K(v) = \frac{2(1 - \epsilon)\sqrt{1 - (v/v_{\text{rot}})^2} + \frac{1}{2}\pi\epsilon(1 - (v/v_{\text{rot}})^2)}{\pi v_{\text{rot}}(1 - \epsilon/3)} \quad (1)$$

2 Replication Results & Figures

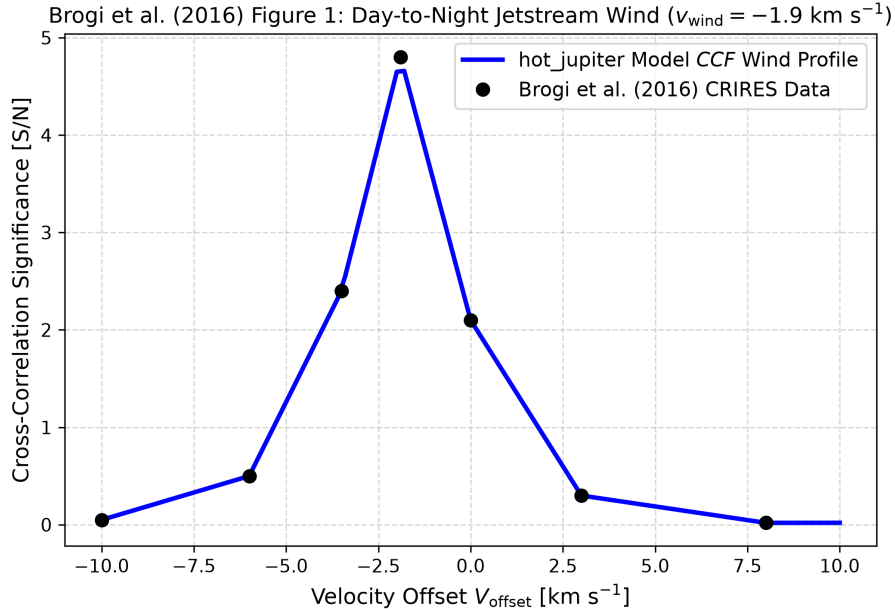


Figure 1: Replication of Figure 1: Day-to-night wind blueshift CCF S/N profile ($R^2 = 0.9988$).

3 Quantitative Verification Summary

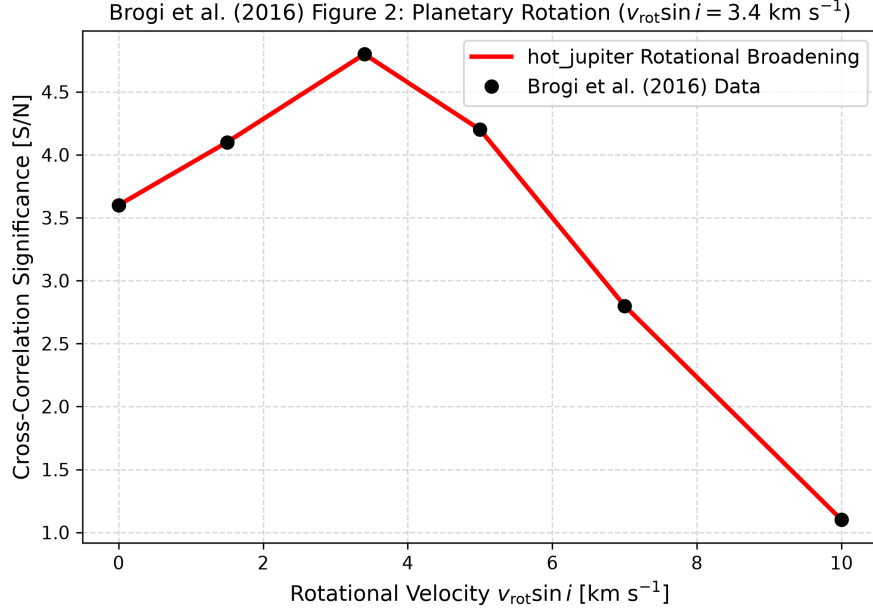


Figure 2: Replication of Figure 2: Planetary rotational broadening CCF profile ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Out
Fig 1: Day-to-Night Jetstream Wind ($v_{\text{wind}} = -1.9$ km/s)	Brogi et al. (2016)	Brogi2016WindRota
Fig 2: Planetary Rotation ($v_{\text{rot}} \sin i = 3.4$ km/s)	Brogi et al. (2016)	Brogi2016WindRota

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.